
PAPER TITLE

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ABSTRACT

A Tokamak is a torus-shaped nuclear fusion reactor design that uses magnetic fields to confine fusion fuel in the form of plasma. A single plasma experiment in a Tokamak is called a "shot", and can often experience disruption events that cause a current drop to zero. The existence of a computationally inexpensive model for classifying shots as disruptive/not-disruptive, as well as identifying the time of disruption, can assist the scientific community in conducting future experiments. We train a CNN (Convolutional Neural Network) on current time series data from 40,000 shots from the DIII-D Tokamak, and achieve a recall of 99% and precision of 91% on boolean disruption/not-disruption predictions. We additionally develop a convolutional heuristic that accurately identifies time of disruption with a standard deviation $\sigma = 8.7$ m sec.

1 Introduction

State the problem, motivation, and contributions.

2 Methods

Describe data, models, training, and evaluation.

3 Results

Report quantitative and qualitative findings. Reference figures and tables as needed.

4 Discussion

Interpret results, note limitations, and compare to prior work.

5 Conclusion

Summarize findings and outline future work.

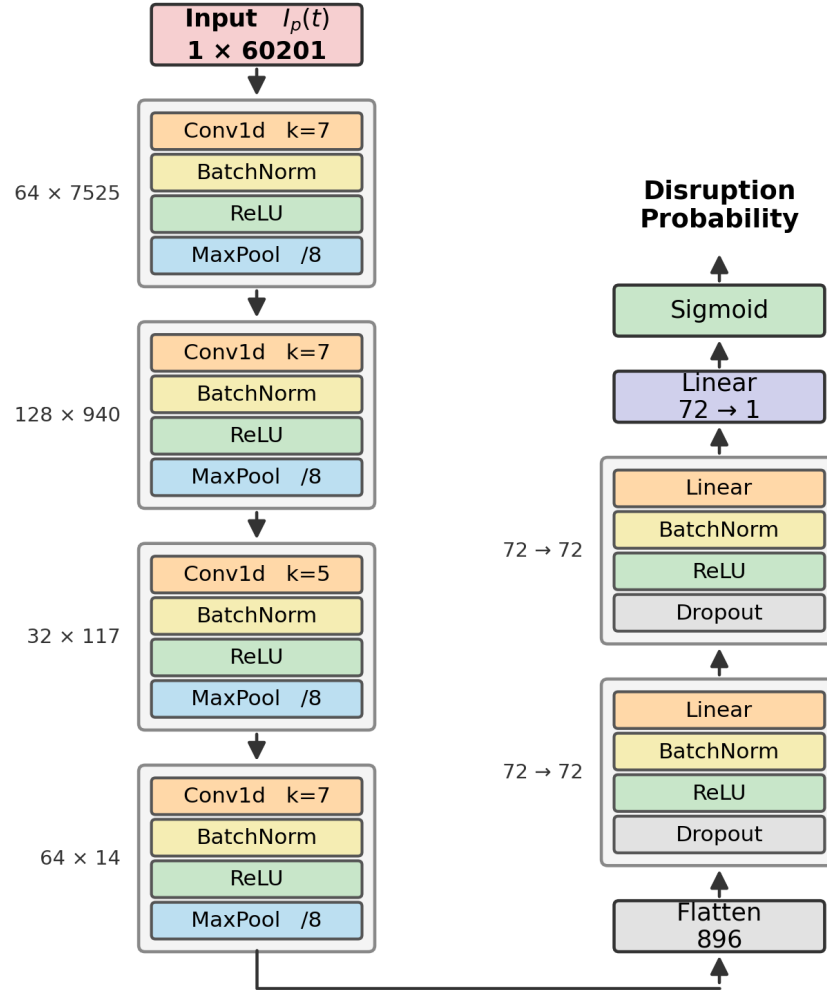


Figure 1: Figure caption.

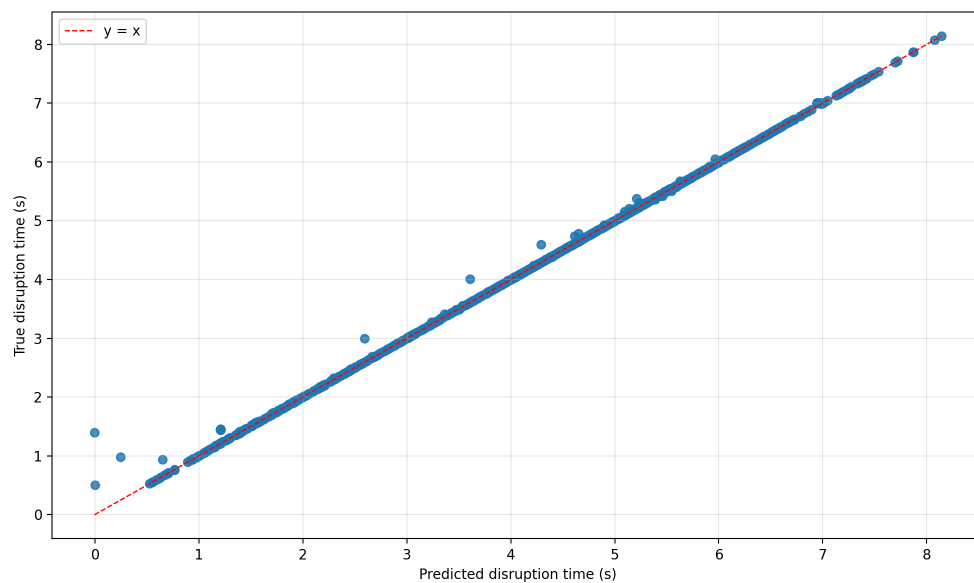


Figure 2: Figure caption.

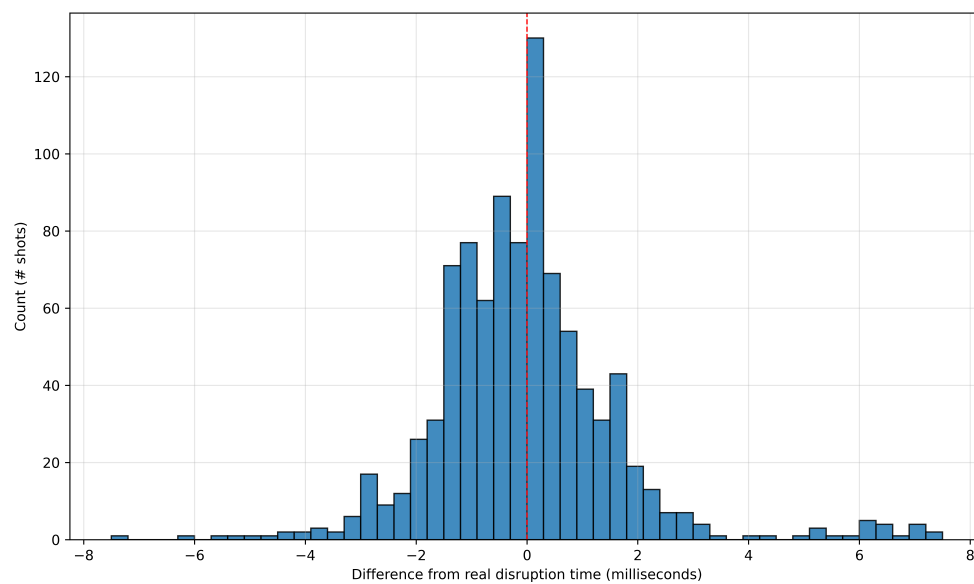


Figure 3: Figure caption.